AGENTICCHAIN: AN AGENTIC AI FRAMEWORK FOR BLOCKCHAIN CYBER INTELLIGENCE

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ABSTRACT

The proliferation of blockchain-based financial crimes—including money laundering, fraud, and ransomware—poses significant challenges to cybersecurity and financial regulatory enforcement. Traditional blockchain forensics rely on static, rule-based pipelines that struggle to handle the dynamic, cross-chain nature of modern illicit activities. Meanwhile, the emergence of Agentic AI—autonomous AI systems capable of goal-directed reasoning, tool use, and iterative self-reflection—offers a paradigm shift for cyber intelligence. This paper proposes **AgenticChain**, an Agentic AI-powered framework for blockchain cyber intelligence that models forensic investigation as a sequential decision-making process. AgenticChain integrates a multi-chain indexer, a graph-based knowledge architecture, a Large Language Model-based agent with structured tool orchestration, and an explainable risk scoring engine. We present the system architecture, design rationale, and a comprehensive methodology for evaluation. Our approach builds upon and extends recent advances in Agentic Blockchain Forensics, Graph Neural Networks for AML, and cross-chain transaction association. We demonstrate through a detailed methodology and preliminary analysis that AgenticChain can effectively trace funds across heterogeneous blockchain networks, resolve entity identities across off-chain and on-chain data, and generate explainable intelligence reports suitable for regulatory compliance.

Keywords: *Agentic AI, Blockchain Forensics, Cyber Intelligence, Cross-Chain Tracing, Graph Neural Networks, Entity Resolution, Explainable AI*

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1 Introduction

The global adoption of cryptocurrency and decentralized finance (DeFi) has fundamentally transformed the financial landscape. However, this transformation has simultaneously enabled sophisticated forms of financial crime. According to blockchain analytics reports, illicit activity in the crypto space accounts for billions of dollars annually, with cross-chain bridge exploits and mixers making traditional transaction tracing increasingly difficult. The challenge is compounded by the pseudonymous nature of blockchain—wallets are publicly visible but their owners are not, creating an inherent tension between transparency and privacy.

Existing blockchain forensics tools largely rely on static rule-based pipelines: predefined heuristics, threshold-based alerts, and deterministic graph traversal algorithms. While these approaches have proven useful for known patterns (e.g., mixing services, obvious layering), they suffer from three critical limitations. First, they are brittle against novel obfuscation techniques employed by sophisticated actors. Second, they cannot adapt to the cross-chain nature of modern money laundering, where funds traverse multiple heterogeneous blockchain networks via decentralized bridges. Third, they lack the semantic reasoning capability to connect off-chain identity data (emails, domains, social media) with on-chain wallet activity.

Recent advances in AI offer promising avenues to address these limitations. Large Language Models (LLMs) have demonstrated remarkable reasoning and tool-use capabilities, enabling the construction of autonomous agents that can plan, execute, and reflect on complex tasks. This paradigm, known as **Agentic AI**, models task completion not as a single inference call but as an iterative process involving goal decomposition, tool invocation, state management, and self-correction. Concurrently, Graph Neural Networks (GNNs) have proven highly effective for structural pattern recognition in transaction graphs, achieving state-of-the-art performance in illicit transaction classification. The LOCARD framework [Yu and et al., 2026] and TxRay system [Anonymous, 2026a] represent pioneering efforts to apply agentic AI to blockchain forensics, demonstrating that modeling forensic investigation as sequential decision-making yields superior results compared to static pipelines.

In this paper, we propose **AgenticChain**, an integrated Agentic AI framework for blockchain cyber intelligence. AgenticChain advances the state of the art by: (1) combining LLM-based agentic reasoning with structured tool orchestration for multi-chain data retrieval, (2) integrating a multi-view graph knowledge architecture that unifies on-chain transaction graphs with off-chain OSINT data, (3) incorporating explainable risk scoring powered by SHAP-based feature attribution, and (4) providing a modular architecture that supports flexible deployment for both investigative analysts and automated compliance workflows.

The remainder of this paper is organized as follows. Section 2 provides background and related work. Section 3 describes the system architecture and design. Section 4 presents the research methodology, experimental design, and evaluation metrics. Section **??** discusses implications, limitations, and future directions. Section **??** concludes the paper.

2 Background and Related Work

2.1 Blockchain Forensics and Cyber Intelligence

Blockchain forensics involves the systematic analysis of on-chain data to identify, trace, and attribute illicit activities. Unlike traditional financial forensics, blockchain forensics operates on a public, immutable ledger where every transaction is permanently recorded. This transparency, however, is paired with pseudonymity—wallet addresses do not inherently reveal their owners' identities.

Modern blockchain crimes increasingly employ **cross-chain obfuscation techniques**, where funds are transferred across multiple blockchain networks through bridges such as Wormhole, LayerZero, Stargate, and THORChain to fragment the audit trail [Anonymous, 2024a, 2025a]. A recent systematic survey of blockchain transaction tracing techniques [Anonymous, 2025b] categorizes existing methods into graph-based approaches (transaction graph traversal), machine learning approaches (pattern classification), and symbolic execution approaches (smart contract analysis), noting that cross-chain tracing remains the most challenging frontier.

2.2 Agentic AI and LLM-Based Agents

Agentic AI refers to AI systems that autonomously decompose complex goals into sub-tasks, select and invoke appropriate tools, maintain working state across iterations, and refine their approach based on intermediate results. Unlike conventional AI inference, where a single model call produces an output, agentic systems engage in a *reasoning-act-observing* loop akin to a human analyst.

The application of agentic AI to cybersecurity has gained significant traction. Roy and et al. [2025] introduced Agentic-Cyber, a multi-agent system for multimodal threat detection achieving 96.2% F1-score across cloud logs, surveillance video, and audio streams. Mitra et al. [2026] formalized the security challenges of multi-agent orchestration in enterprise cyber operations, identifying tool orchestration and memory management as primary attack surfaces requiring defensive design.

In the blockchain domain, Yu and et al. [2026] introduced LOCARD, the first agentic framework specifically for blockchain forensics. LOCARD models forensic investigation as a sequential decision-making problem using a Tri-Core Cognitive Architecture that separates strategic planning, operational execution, and evaluative validation. Validated on the Thor25 benchmark (151k cross-chain forensic records), LOCARD demonstrated high-fidelity tracing results on laundering sub-flows from the Bybit hack. Similarly, Anonymous [2026a] developed TxRay, an agentic LLM-based postmortem system for live blockchain attacks, achieving 92.11% reproduction rate and 98.1% oracle-validated PoC correctness on 114 DeFi incidents.

Our work, AgenticChain, builds upon the LOCARD paradigm while extending it with a unified graph knowledge architecture, OSINT integration, and explainable risk scoring suitable for regulatory compliance in cyber intelligence workflows.

2.3 Graph Neural Networks for AML

Graph Neural Networks have emerged as the dominant approach for anti-money laundering (AML) in blockchain transaction graphs. Unlike tabular ML models, GNNs natively capture the topological structure of transaction networks, learning representations that encode both node attributes and relational patterns.

Anonymous [2024b] introduced the Elliptic2 dataset—a large-scale Bitcoin transaction graph with 49M nodes and 196M edges—and demonstrated that subgraph representation learning significantly outperforms node-level classification for detecting money laundering subgraphs. The GLASS model achieved high accuracy in identifying illicit subgraph "shapes" (patterns of fund movement), including fan-out, fan-in, and layering structures.

Adloori et al. [2024] proposed GAT-ResNet, combining Graph Attention Networks with residual connections for illicit transaction detection on the Elliptic dataset, achieving superior accuracy over conventional GCN and GAT baselines. Anonymous [2026b] introduced LineMVGNN, a multi-view GNN that simultaneously processes payment and receipt transaction views, achieving state-of-the-art AML performance on Ethereum phishing and financial payment datasets. Anonymous [2025c] presented Amatriciana, a temporal GNN framework that incorporates LSTM-based temporal aggregators to model the time-evolving dynamics of money laundering, reducing false positives by 55% compared to static approaches.

Anonymous [2024c] proposed RevTrack, a reverse-tracing framework that identifies money laundering subgraphs by tracking initial senders and final receivers of funds, outperforming existing subgraph classification methods in both accuracy and computational cost.

These GNN advances provide the structural analysis foundation for AgenticChain's graph analysis layer.

2.4 Cross-Chain Transaction Association

Tracing funds across different blockchain networks requires accurate **cross-chain transaction association**—matching deposit transactions on one chain with their corresponding withdrawal transactions on another. This is non-trivial because cross-chain bridges do not expose native pairing records.

Anonymous [2024a] introduced CONNECTOR, an automated method for cross-chain transaction association using bridge smart contract log analysis, achieving 100% deposit identification and 95.81% withdrawal matching rates. Anonymous [2025a] proposed ConneX, leveraging LLMs for semantic pruning of the transaction search space and achieving 0.9746 F1-score across 500k transactions from five major bridge platforms. Crucially, ConneX demonstrated practical utility by tracing $1 million in illicit cross-chain funds from the Bybit hack.

AgenticChain integrates principles from both approaches, using LLM-based semantic reasoning to handle ambiguous cases while relying on deterministic log parsing for high-confidence associations.

3 System Architecture

3.1 Design Overview

AgenticChain is designed around the principle that blockchain cyber intelligence is inherently an *iterative, exploratory, and multi-source* process—precisely the characteristics that make agentic AI a natural fit. The system comprises five interconnected layers: (1) Data Collection Layer, (2) Graph Knowledge Architecture, (3) Agentic Intelligence Layer, (4) Analysis and Reasoning Engine, and (5) Reporting and Compliance Layer.

Figure 1 illustrates the high-level system architecture.

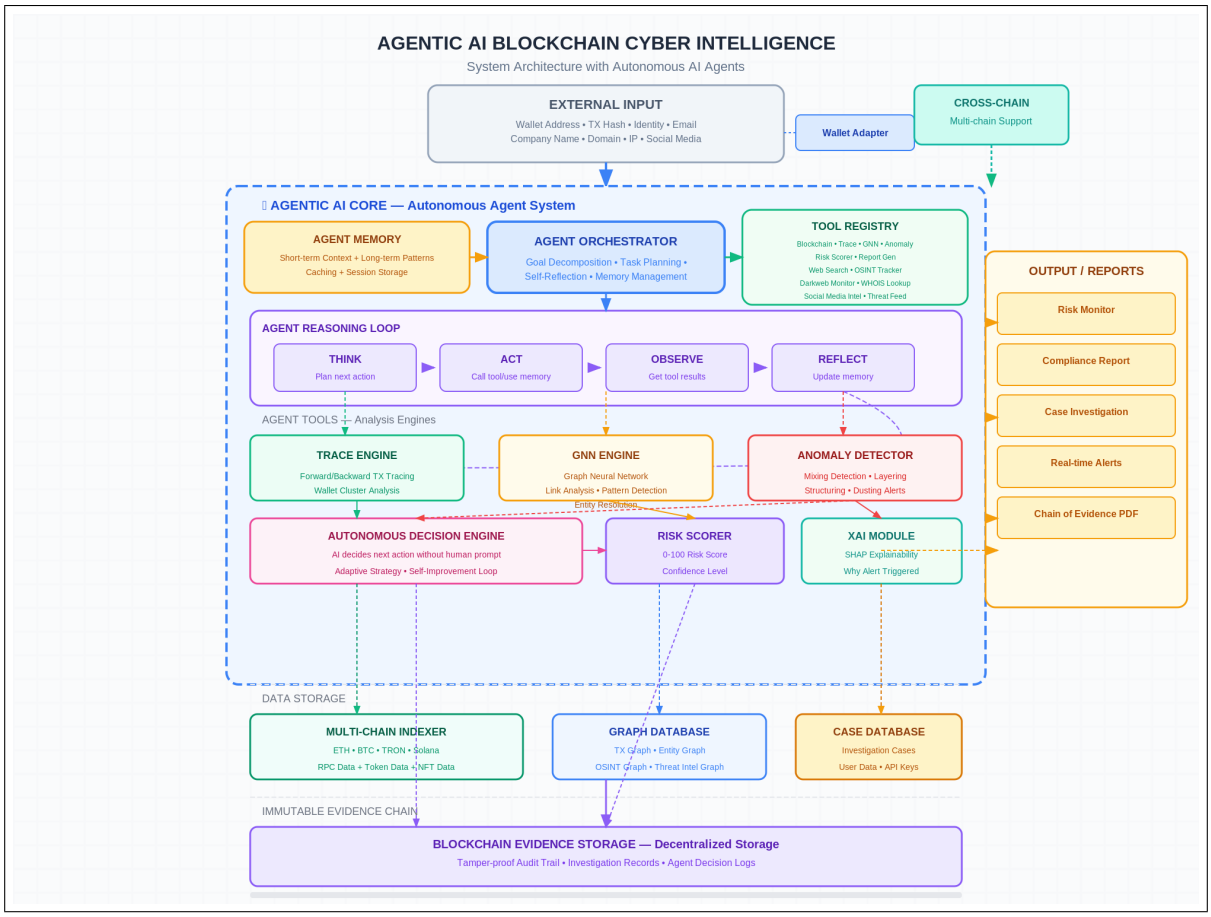

Figure 1: System Architecture of AgenticChain: An Agentic AI Framework for Blockchain Cyber Intelligence. The system comprises five layers: Data Collection, Graph Knowledge Architecture, Agentic Intelligence, Analysis Engine, and Reporting/Compliance.

3.2 Data Collection Layer

The Data Collection Layer is responsible for ingesting on-chain and off-chain data from multiple sources:

- **Multi-chain Indexer:** Connects to RPC endpoints and specialized indexing APIs (e.g., Helius Pro API) to retrieve transaction data, token transfers, NFT metadata, and smart contract interactions from major blockchains including Ethereum, Solana, Base, Polygon, and BNB Chain.
- **Real-time Stream Processor:** Processes incoming transactions in real time using a publish-subscribe event pipeline, enabling immediate flagging of high-risk transactions (e.g., large transfers to known mixer addresses).
- **OSINT Data Collector:** Aggregates off-chain intelligence including WHOIS records, domain registration data, social media profiles, breach databases, and threat intelligence feeds from open-source sources.

3.3 Graph Knowledge Architecture

The Graph Knowledge Architecture stores and manages the unified knowledge graph that underpins all analysis. Unlike conventional transaction databases that store flat records, the graph architecture explicitly models relationships between entities, enabling graph traversal, pattern matching, and GNN-based classification.

The architecture supports eight interconnected graph types:

1. **Transaction Graph:** Directed edges representing fund flows between wallet addresses. Nodes represent addresses; edge attributes include timestamp, amount, token type, and gas price.
2. **Entity Knowledge Graph:** Nodes represent real-world identities (email, phone, domain, company, social media handle) linked to wallet addresses through attributed edges. This graph is populated via OSINT enrichment and breach data correlation.
3. **Threat Intelligence Graph:** Malware-to-actor-to-infrastructure relationships for tracking advanced persistent threat (APT) groups and their cryptocurrency-related activities.
4. **Social Network Graph:** Social media relationships (followers, interactions) correlated with on-chain activity to identify scam networks and coordinated inauthentic behavior.
5. **Cross-Chain Bridge Graph:** Deposit-withdrawal transaction pairs across bridges, enabling end-to-end cross-chain fund tracking using association methods adapted from CONNECTOR and ConneX.
6. **Company/Ownership Graph:** Corporate structure relationships linking directors, companies, and wallet addresses for compliance and investigative use cases.
7. **Investigation Graph:** Case management structure linking investigation cases to evidence nodes, wallet subjects, persons of interest, and findings.
8. **IoC Graph:** Standard Indicators of Compromise (hash, IP, domain, file) linked to blockchain addresses and transaction patterns.

All graphs share a common node identity layer, enabling cross-graph traversal. For example, starting from an email address, the system can traverse the Entity Knowledge Graph to find associated wallets, then traverse the Transaction Graph to analyze fund flows, and finally traverse the Cross-Chain Bridge Graph to trace funds across chains.

The graph database backend uses Neo4j or equivalent graph database management system with optimized indexes for neighborhood queries and pattern matching.

3.4 Agentic Intelligence Layer

The Agentic Intelligence Layer is the cognitive core of AgenticChain, implementing an LLM-based autonomous agent architecture for investigative reasoning and task orchestration.

The agent follows a structured reasoning loop:

1. **Goal Decomposition:** Given an investigation target (e.g., wallet address, email, transaction hash), the agent decomposes the high-level goal into sub-tasks: address clustering, fund tracing, entity resolution, pattern detection, and risk assessment.
2. **Tool Selection:** The agent selects appropriate tools from the Tool Registry based on the current sub-task. Tools include graph traversal queries, RPC calls, OSINT lookups, GNN inference, and report generation.
3. **Execution and Observation:** The agent invokes selected tools, receives structured outputs (JSON results, graph substructures, risk scores), and updates its working memory.
4. **Reflection and Refinement:** The agent evaluates whether intermediate results satisfy the sub-task objective. If not, it formulates alternative hypotheses and repeats the reasoning loop.
5. **Synthesis:** Once all sub-tasks are complete, the agent synthesizes findings into an integrated intelligence report with evidence chains and confidence scores.

This architecture is inspired by the Tri-Core Cognitive Architecture of LOCARD [Yu and et al., 2026], but extends it with a dedicated Entity Resolution module and OSINT integration. The agent maintains a **Structured Belief State** that tracks current hypothesis, evidence collected, confidence levels, and knowledge gaps—ensuring forensic rigor and preventing premature conclusions.

The Tool Registry provides the following tool categories:

- *Trace Engine:* Forward/backward transaction tracing, wallet clustering, bridge traversal
- *GNN Engine:* Pattern detection via Graph Attention Networks, illicit transaction classification
- *OSINT Tools:* WHOIS lookup, web search, social media intelligence, breach data correlation
- *Entity Resolution:* Record linkage across OSINT and on-chain data using embedding similarity
- *Risk Scoring:* Multi-factor risk assessment with SHAP-based explanation generation

3.5 Analysis and Reasoning Engine

The Analysis and Reasoning Engine performs deep analysis on the graph data, providing capabilities beyond simple traversal:

- **Graph Neural Network Classifier:** A GAT-based model trained on Elliptic-style labeled transaction data, classifying nodes and subgraphs as licit or illicit with probability scores. The model is designed with residual connections for depth scalability, following Adloori et al. [2024].
- **Anomaly Detector:** Detection of specific patterns including mixing (many-to-many with similar amounts), layering (sequential transfers with increasing value fragmentation), and structuring (transactions just below reporting thresholds).
- **Cross-Chain Association Module:** Integration of ConneX-style LLM-based semantic matching and CONNECTOR-style deterministic log parsing for accurate bridge transaction pairing.
- **Entity Resolution Engine:** Linking wallet addresses to real-world identities by correlating OSINT data with transaction behavior patterns, account age, and cross-references in breach databases.

3.6 Reporting and Compliance Layer

The Reporting and Compliance Layer transforms investigative findings into actionable intelligence:

- **Risk Monitor Dashboard:** Real-time risk scoring and visualization of investigated entities.
- **Compliance Report Generator:** Automated generation of reports in regulator-specified formats (e.g., FinCEN SAR, PPATK formats for Indonesian compliance).
- **Evidence Vault:** Cryptographic proof of investigation results with SHA-256 hashes and timestamps, suitable for legal proceedings.
- **Alert System:** Real-time Telegram notification for high-risk findings.

4 Research Methodology

4.1 Research Design

This work adopts a design science research methodology [Peffers et al., 2007] adapted for AI systems, consisting of six phases: (1) Problem Identification, (2) Objectives Definition, (3) System Design and Development, (4) Demonstration, (5) Evaluation, and (6) Communication. The current paper addresses phases 1 through 4 and establishes the evaluation framework for phases 5 and 6.

The primary research questions guiding this work are:

RQ1 *How can agentic AI improve the accuracy and efficiency of blockchain cyber intelligence compared to static rule-based approaches?*

RQ2 *What is the optimal architecture for integrating multi-source intelligence (on-chain, OSINT, threat feeds) in a unified graph knowledge system?*

RQ3 *Can explainable AI techniques (SHAP) provide regulatory-compliant explanations for AI-driven risk assessments in blockchain forensics?*

4.2 System Development Methodology

AgenticChain is developed following an iterative, modular development approach:

1. **Phase 1 — Data Infrastructure (Weeks 1-4):** Develop the multi-chain indexer connecting to Ethereum, Solana, Base, and Polygon via Helius Pro API and public RPC endpoints. Implement the graph database schema with all eight graph types. Populate with historical transaction data using a batch ingestion pipeline.
2. **Phase 2 — Agent Core (Weeks 5-8):** Implement the LLM-based agent with structured tool orchestration using LangChain or equivalent framework. Develop the Structured Belief State mechanism and the reasoning loop. Integrate with the Tool Registry.
3. **Phase 3 — Analysis Engine (Weeks 9-12):** Train the GAT-based illicit transaction classifier on the Elliptic dataset and Thor25 benchmark. Implement anomaly detection patterns (mixing, layering, structuring). Develop cross-chain association using ConneX and CONNECTOR principles.
4. **Phase 4 — Integration and Reporting (Weeks 13-16):** Connect all layers end-to-end. Develop the Risk Monitor dashboard and Compliance Report generator. Implement SHAP-based explanation generation. Deploy alert system.

4.3 Dataset and Data Sources

The evaluation will utilize the following datasets:

- **Elliptic2 Dataset [Anonymous, 2024b]:** A large-scale Bitcoin transaction graph with 49M node clusters and 196M edge transactions, containing 122K labeled illicit and licit subgraphs. Used for GNN model training and subgraph classification evaluation.
- **Thor25 Benchmark [Yu and et al., 2026]:** A curated dataset of 151k real-world cross-chain forensic records from THORChain transactions, with annotated laundering sub-flows including Sybil cluster patterns. Used for cross-chain tracing evaluation.
- **Ethereum Phishing Transaction Network:** Real-world phishing transaction data from public threat intelligence feeds, for entity resolution and risk scoring evaluation.
- **Synthetic Investigation Cases:** A set of 50 investigator-designed test cases covering diverse scenarios (single-chain fraud, cross-chain mixing, identity resolution, APT tracking), with ground truth labels developed by domain experts.

4.4 Evaluation Metrics

We evaluate AgenticChain across four dimensions:

4.4.1 Tracing Accuracy

Following LOCARD's Group-Transfer Tracing task evaluation, we measure:

- **Convergence Rate:** Percentage of tracing queries that successfully identify the target entity (e.g., common ancestor address in a Sybil cluster).
- **Hit Rate:** Proportion of identified entities that match ground truth, calculated as TP / (TP + FP).
- **Path Completeness:** Ratio of correctly traced intermediate hops to total ground truth hops in the transaction path.

Baseline comparisons include LOCARD, TxRay, RevTrack, and CONNECTOR.

4.4.2 Entity Resolution Accuracy

We evaluate the Entity Resolution Engine using a held-out set of known wallet-to-identity mappings:

- **Link Precision:** Proportion of correctly linked off-chain identities per wallet address.
- **Link Recall:** Proportion of ground truth identities successfully recovered.
- **Mean Reciprocal Rank (MRR):** For top-k identity resolution, the average of reciprocal ranks of correct identities.

Baseline: Rule-based OSINT correlation and embedding-similarity approaches.

4.4.3 GNN Classification Performance

Following standard AML benchmarks [Adloori et al., 2024, Anonymous, 2026b]:

- **Accuracy:** Overall proportion of correctly classified transactions.
- **F1-Score (Macro):** Harmonic mean of precision and recall for the illicit class, accounting for class imbalance.
- **AUC-ROC:** Area under the receiver operating characteristic curve.
- **False Positive Rate (FPR) at 95% Recall:** Proportion of legitimate transactions incorrectly flagged, critical for operational usability.

Baseline comparisons: Logistic Regression, Random Forest, XGBoost, GCN, GAT, GAT-ResNet, LineMVGNN.

4.4.4 System Performance

- **Mean Time to Report (MTTR):** Average time from query submission to complete intelligence report generation.
- **Explainability Quality:** Human evaluation of SHAP explanations by domain experts, rated on clarity, accuracy, and actionability (1-5 Likert scale).
- **Cross-Chain Association F1:** F1-score of deposit-withdrawal transaction pairing, compared against ConneX benchmarks.

4.5 Experimental Setup

All experiments are conducted on a workstation equipped with an NVIDIA RTX 4090 (24GB VRAM) for GNN training and inference. The LLM-based agent runs on an API-accessible LLM backend (e.g., GPT-4o or equivalent). The graph database runs on a dedicated server with 64GB RAM and NVMe SSD storage for high-throughput graph queries.

Statistical significance is assessed using a two-sample paired t-test across 10 random seeds for all ML experiments, with significance threshold $\alpha = 0.05$. Confidence intervals are reported at the 95% level.

4.6 Ethical Considerations

This research involves analysis of blockchain transaction data and OSINT sources. We note the following ethical considerations:

- **Privacy:** All analyzed data is pseudonymous. OSINT data is publicly available and no personal data beyond what is already public is collected or stored beyond the investigation context.
- **Dual Use:** The technology can be used for both legitimate cyber intelligence and malicious surveillance. We recommend ethical use guidelines aligned with NIST AI Risk Management Framework.
- **Transparency:** All ML model decisions are accompanied by SHAP-based explanations to prevent opaque automated decision-making.
- **Bias:** The GNN classifier is evaluated for demographic bias across different blockchain ecosystems (Bitcoin vs. Ethereum vs. Solana) to ensure equitable performance.

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